

Systems Analysis and Design

Chapter 2: Software development life cycle



Learning Objectives

1.1 Choose the right SDLC

1.2 Describe the RUP methodologies

1.3 Describe the Agile methodologies

1.4 The great methodologies debate

68% of all Software Projects fail – ZDNET

1.1 Choose the right SDLC

- McKinsey – 17% of large IT Projects fail miserably
- Geneca - Large IT Projects run 45% over budget, 7% over time, delivering 56% less value
- 75% Project participants lack confidence in their project



Software Projects fail from around 5 to 47 factors – Alexandria University

1.1 Choose the right SDLC

- Organizational Structure
- Badly Defined Requirements
- Unrealistic or Unarticulated goals
- Inability to handle project complexity
- Sloppy development practices
- Inaccurate estimates



Project failures can be controlled

1.1 Choose the right SDLC

- Delivery dates impact project delivery
- Projects estimations can be made as close as possible
- Risks can be re-assessed, controlled and managed
- Staff can be awarded for long work hours



Unified Software Development Process

1.1 Choose the right SDLC

- **Rational Unified Process**
- Agile Unified Process
- Basic Unified Process
- Enterprise Unified Process
- Essential Unified Process
- Open Unified Process
- Oracle Unified Method
- RUP-Systems Engineering

Agile methodology

1.1 Choose the right SDLC

- Kanban
- Scrum
- XP
- Feature-driven development (FDD)
- Dynamic Systems Development Method (DSDM)
- Crystal
- Lean

What is Rational Unified Process ?

1.2 Describe the RUP methodologies

A software engineering process based on best practices in modern software development

- A disciplined approach to assigning and managing tasks and responsibilities in a development organization
- Focus on high quality software that meets the needs of its end users within a predictable schedule and budget

A process framework that can be tailored to specific organization or project needs



RUP is a methodology for delivering projects in a maximum performance manner

RUP uses an integration of approaches & initiatives

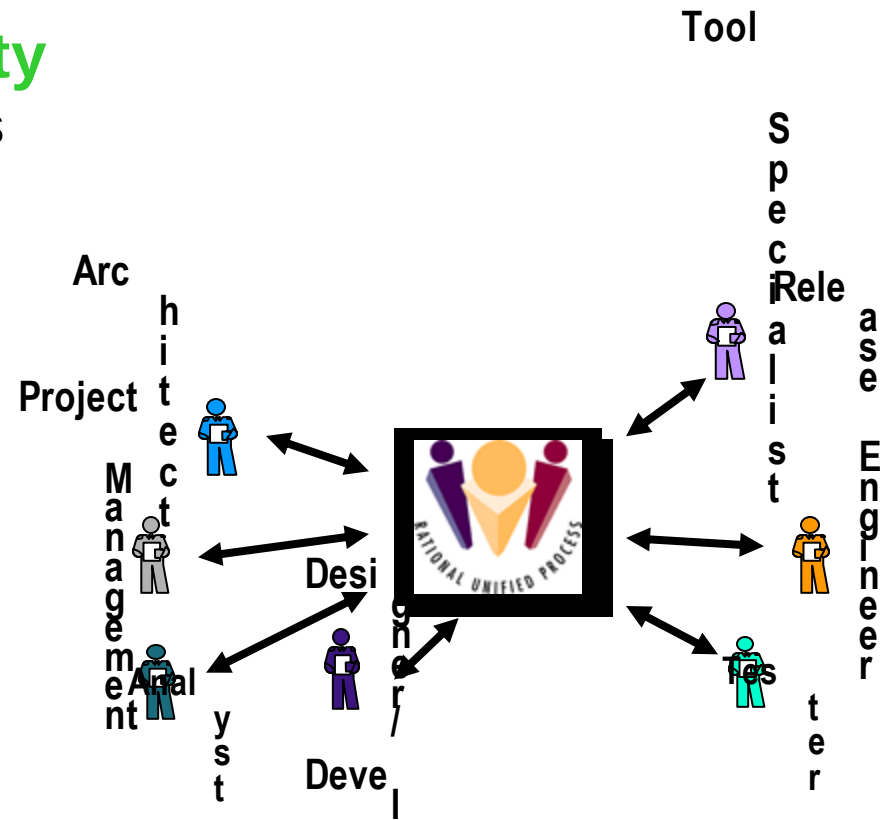
1.2 Describe the RUP methodologies

? Team-Unifying Approach

The RUP unifies a software team by providing a common view of the development process and a shared vision of a common goal

? Increased Team Productivity

- ? knowledge base of all processes
- ? view of how to develop software
- ? modeling language
- ? Rational provides many tools



Key Aspects of RUP

1.2 Describe the RUP methodologies

Risk-driven process

- Risk management integrated into the development process
- Iterations are planned based on high priority risks

Use-Case driven development

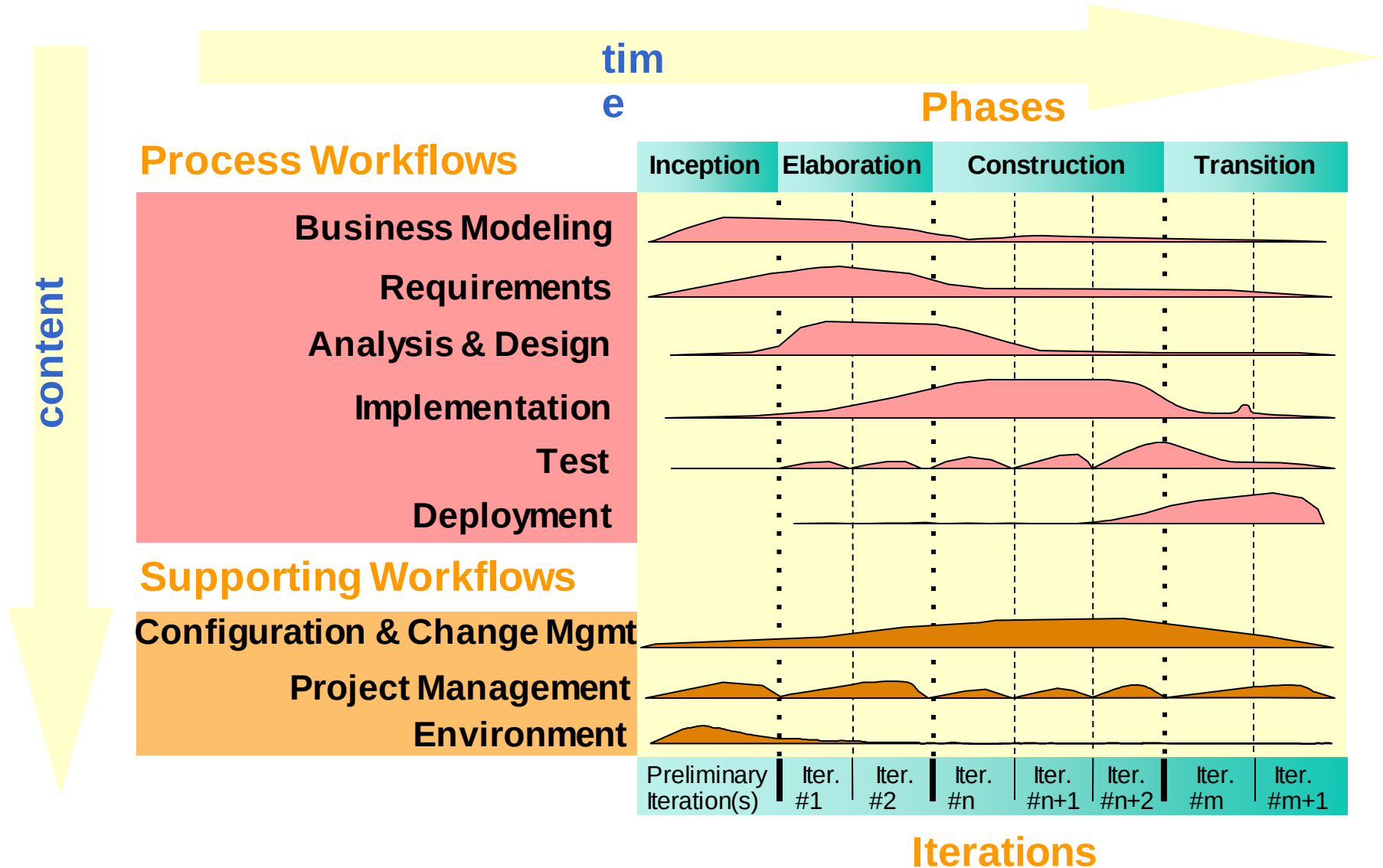
- Use cases express requirements on the system's functionality and model the business as context for the system
- Use cases are defined for the intended system and are used as the basis of the entire development process

Architecture-centric design

- Architecture is the primary artefact to conceptualize, construct, manage, and evolve the system
- Consists of multiple, coordinated views (or models) of the architecture

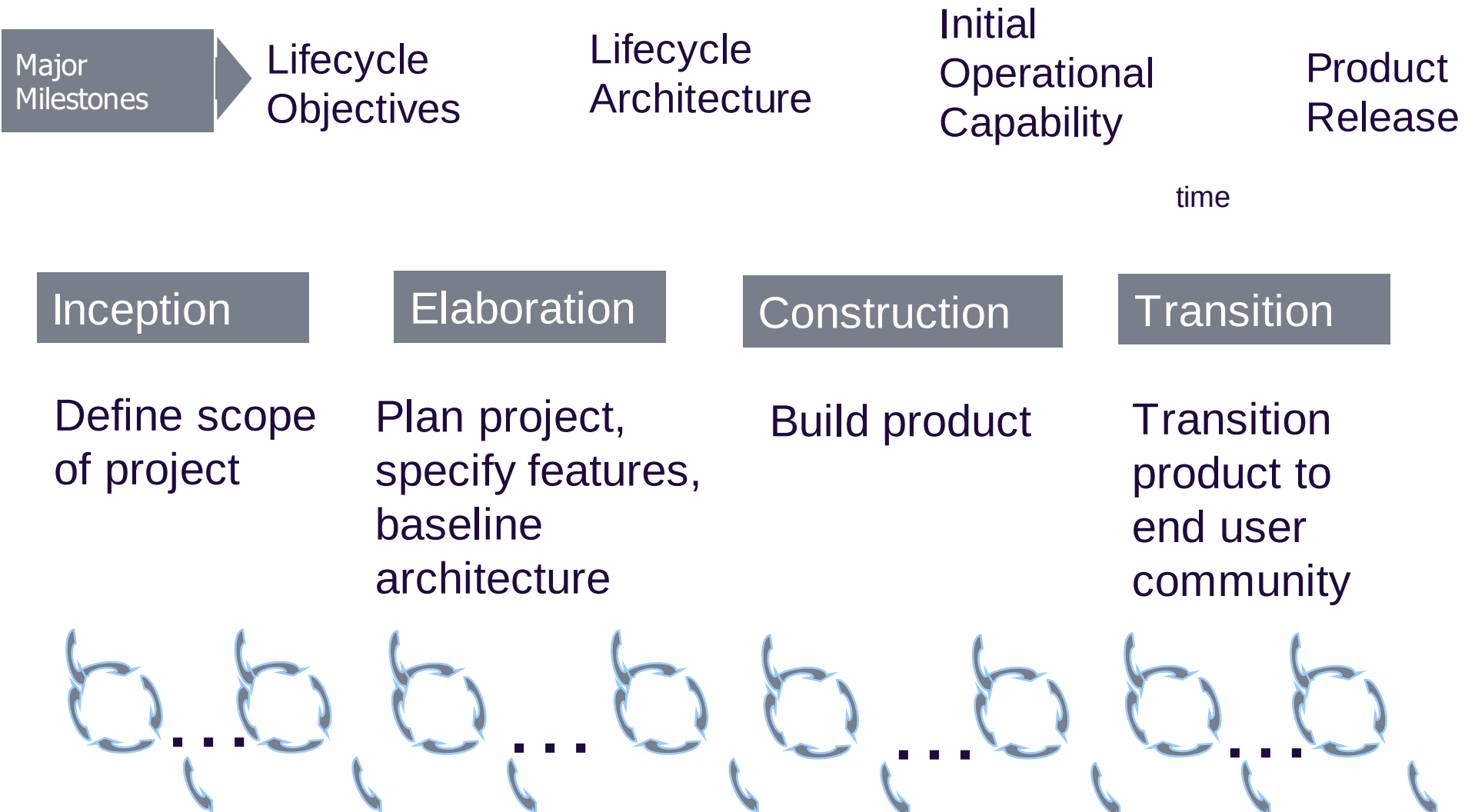
Rational Unified Process – Snapshot

1.2 Describe the RUP methodologies



Dynamic Elements Phases and Milestones

1.2 Describe the RUP methodologies



Inception Phase

1.2 Describe the RUP methodologies

Objective:

- Understand what to build.
 - A vision document:
 - Optional business model
 - An initial project glossary
- Identify key system functionality.
 - A initial use-case model (10% -20%) complete.
- Determine at least one possible solution.
 - One or several prototypes.
- Understand the costs, schedule, and risks associated with the project.
 - An initial risk assessment.
 - Business case
- Decide what process to follow and what tools to use.
 - A project plan

Inception is the first of four RUP phase its all about getting familiar with Project goal and Scope .this phase help you determine the project feasibility , what customer want and how will you get into more resource consumable phase.

Elaboration phase

1.2 Describe the RUP methodologies

Objectives

- Deeper Requirement understanding
 - At least 80% complete use-case model
 - Supplementary requirements capturing
 - non functional requirements
 - None Use case requirement
- Architect consideration.
 - A Software Architecture Description.
 - An executable architectural prototype.
- Risk mitigation and Accurate Cost/Scapulae
 - A revised risk list and a revised business case.
- Development Case refinement
 - A development plan for the overall project
 - coarse-grained project plan
 - showing iterations
 - evaluation criteria for each iteration.

Elaboration is the second of the four phases in the RUP approach. The goal of the Elaboration phase is to define and baseline the architecture of the system in order to provide a stable basis for the bulk of the design and implementation effort in the Construction phase. The architecture evolves out of a consideration of the most significant requirements (those that have a great impact on the architecture of the system) and an assessment of risks.

Construction Phase

1.2 Describe the RUP methodologies

Construction is really about cost-efficient development of a complete product—an operational version of your system—that can be deployed in the user community

Objectives:

- Minimize development costs and achieve some degree of parallelism
- Iteratively develop a complete product that is ready to transition to its user community
- The software product integrated on the adequate platforms.
- The user manuals.
- A description of the current release.

Transition Phase

1.2 Describe the RUP methodologies

The purpose of the transition phase is to transition the software product to the user community. Once the product has been given to the end user, issues usually arise that require you to develop new releases, correct some problems, or finish the features that were postponed.

Objectives:

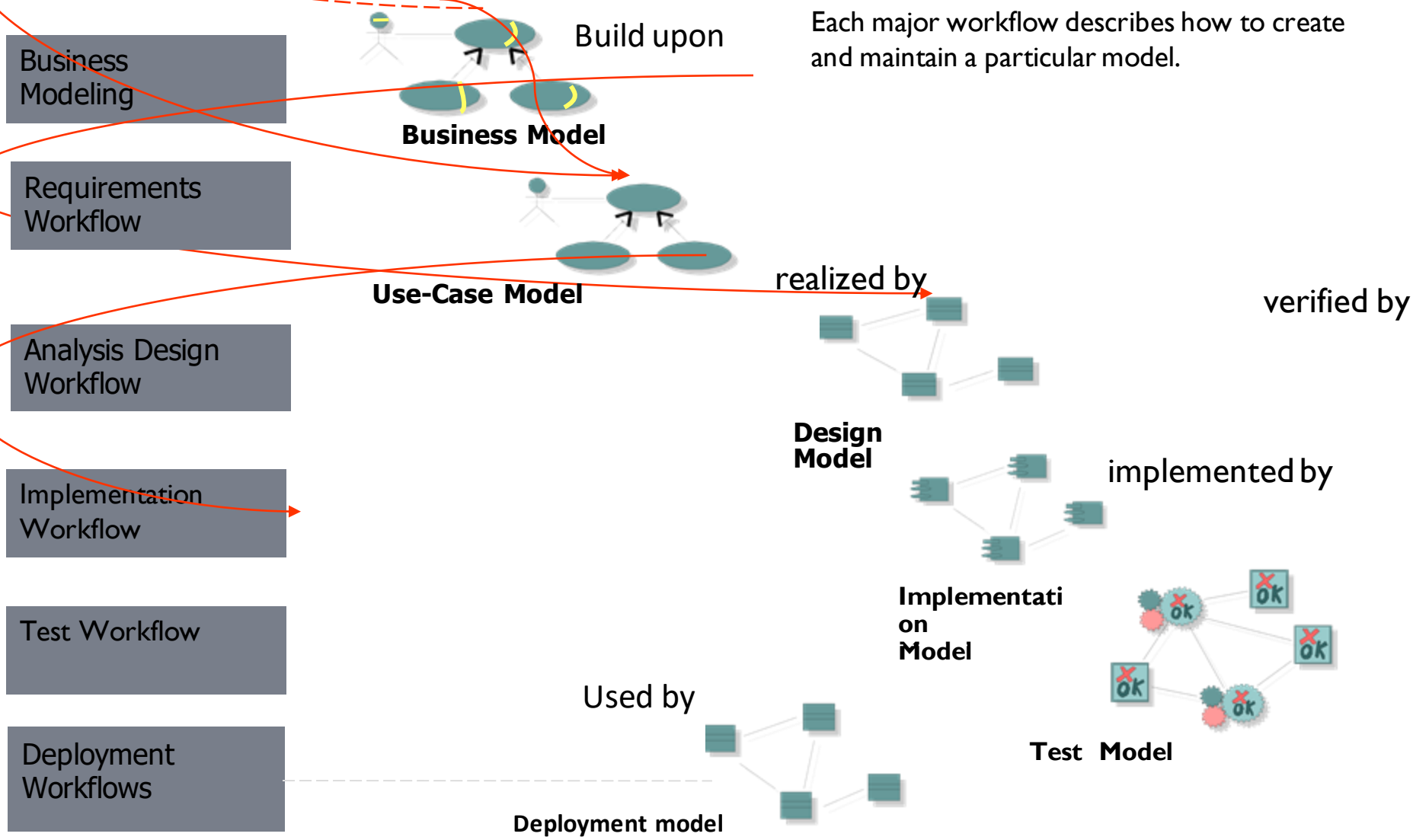
- “beta testing” to validate the new system against user expectations
- parallel operation with a legacy system that it is replacing
- conversion of operational databases
- training of users and maintainers
- roll-out the product to the marketing, distribution, and sales teams
- Improve future project performance through lessons learned

RUP Disciplines

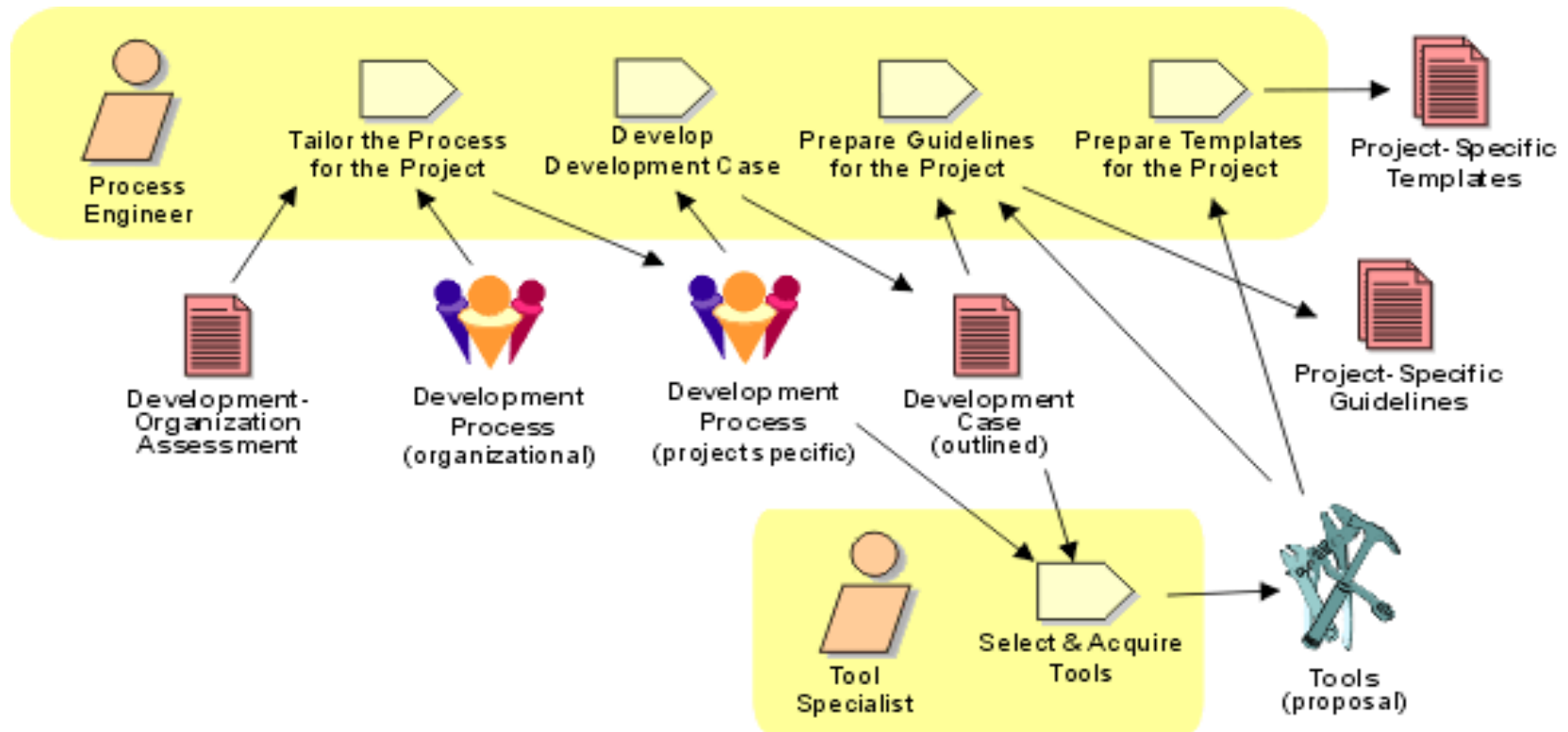
1.2 Describe the RUP methodologies

- In RUP, the process is described at two levels: the discipline level and the workflow detail level. A **Workflow** is a grouping of activities that are often performed "together" to produce a specific result. In particular, workflow details describe groups of activities performed together in a discipline.
- The workflows for the RUP disciplines and workflow details are described using Unified Modeling Language (UML) activity diagrams. Discipline diagrams contain the workflow details of the discipline.

Models and Workflows



Workflow Detail: Prepare Environment for Project



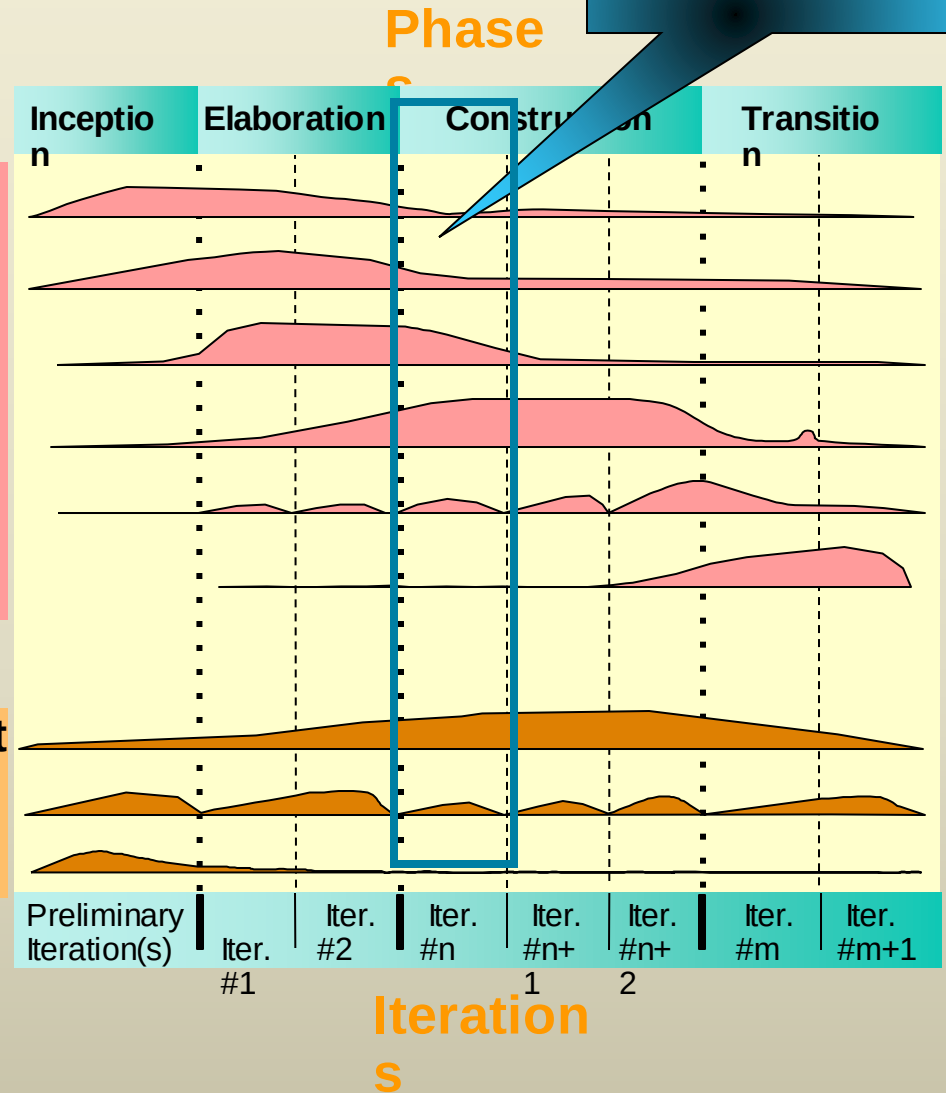
Bringing It All Together...

Process Workflows

Business Modeling
Requirements
Analysis & Design
Implementation
Test
Deployment

Supporting Workflows

Configuration & Change Mgmt
Project Management
Environment



What is Agile?

1.3 Describe the Agile methodologies

- Agile is a time boxed, iterative approach to software delivery that builds software incrementally from the start of the project, instead of trying to deliver it all at once near the end.

incrementally

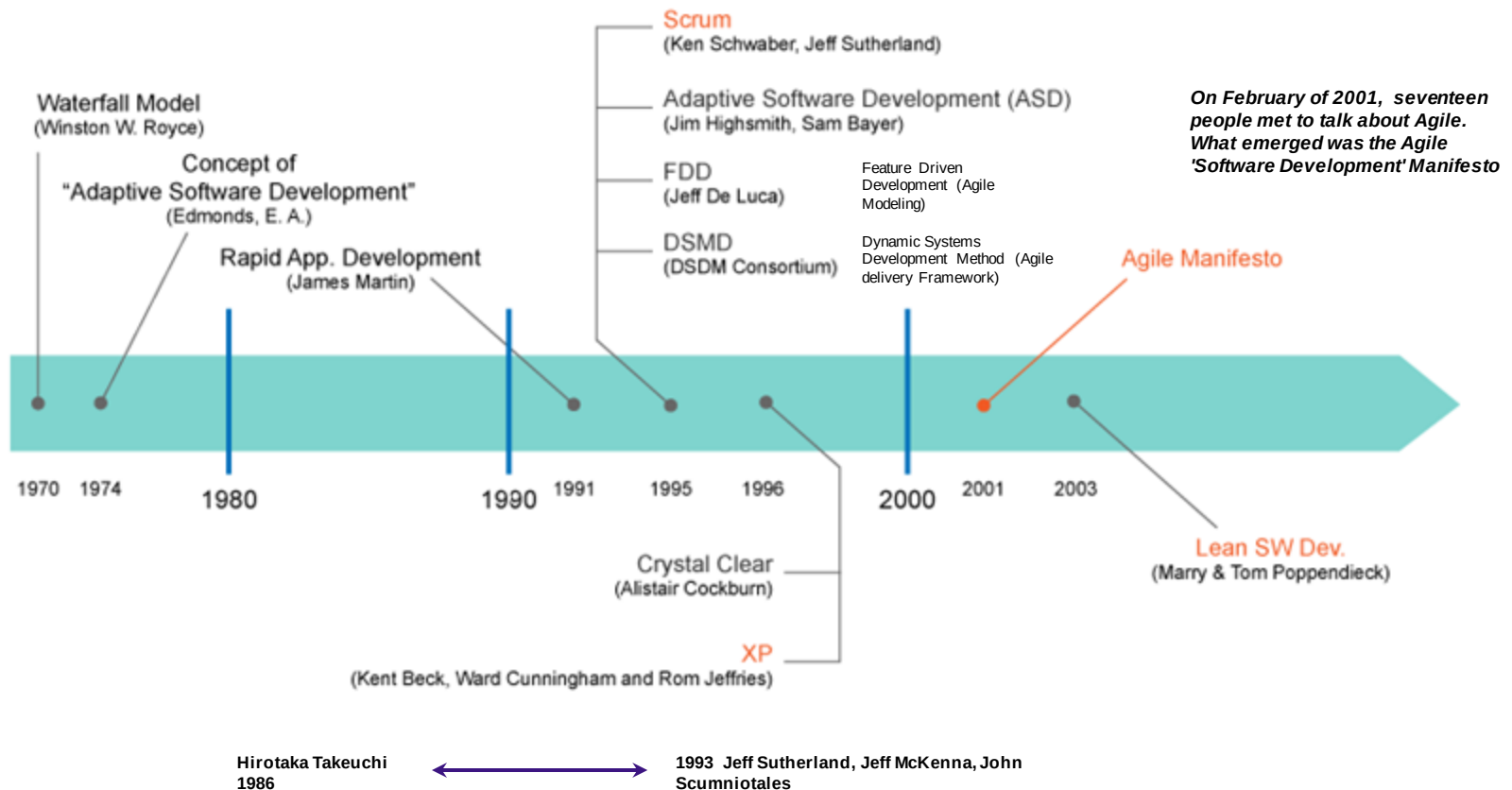


instead of all at once



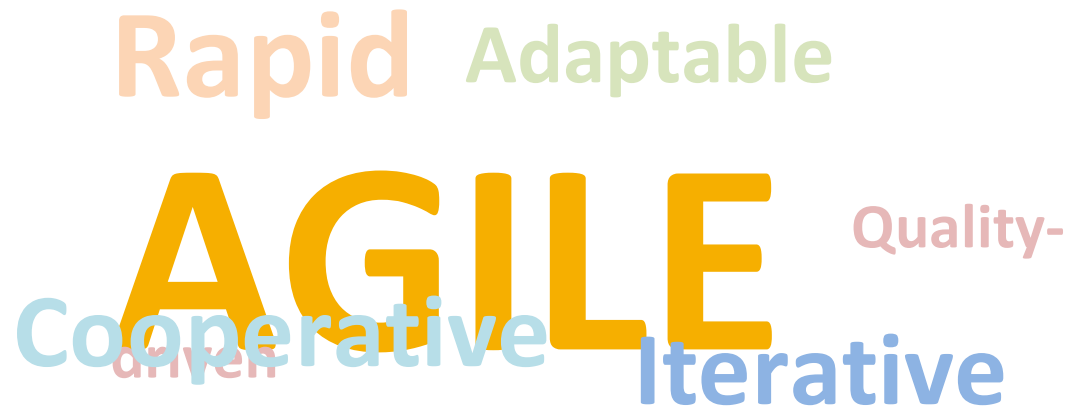
Agile History Timeline

1.3 Describe the Agile methodologies



What is Agile?

1.3 Describe the Agile methodologies



A word cloud centered around the word "AGILE". The word "AGILE" is the largest and most prominent, rendered in a bold, yellow, sans-serif font. Surrounding it are several other words in various colors and sizes, including "Rapid" (orange), "Adaptable" (green), "Quality-" (pink), "Iterative" (blue), "Cooperative" (light blue), and "driven" (small, pink, partially obscured).

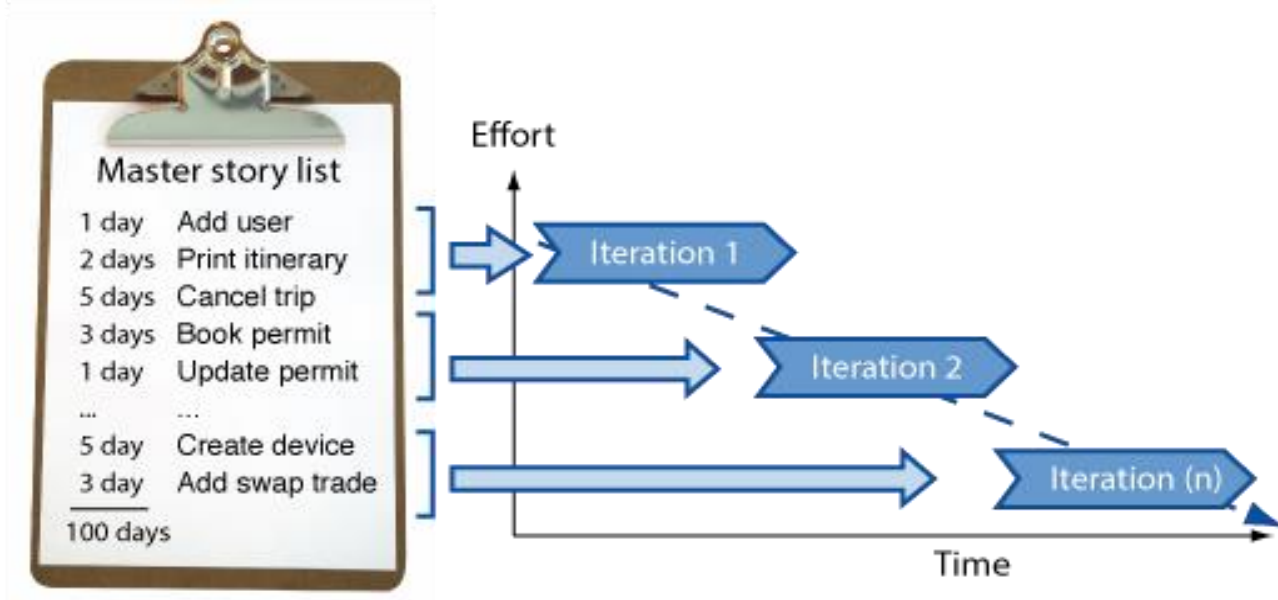
Rapid Adaptable
AGILE
Quality-
Cooperative driven Iterative

Not a process, it's a **philosophy** or **set of values**

What is Agile?

1.3 Describe the Agile methodologies

- It works by breaking projects down into little bits of user functionality called **user stories**, prioritizing them, and then continuously delivering them in short two week cycles called **sprints**.



Modern Agile

1.3 Describe the Agile methodologies

- The industry has become overloaded with techniques, processes, methodologies and tools that have fallen under the Agile umbrella
- Modern Agile aims to move Agile to the next level, not by adding complexity, but by simplifying— stressing only adherence to four principles



Modern Agile guiding principles

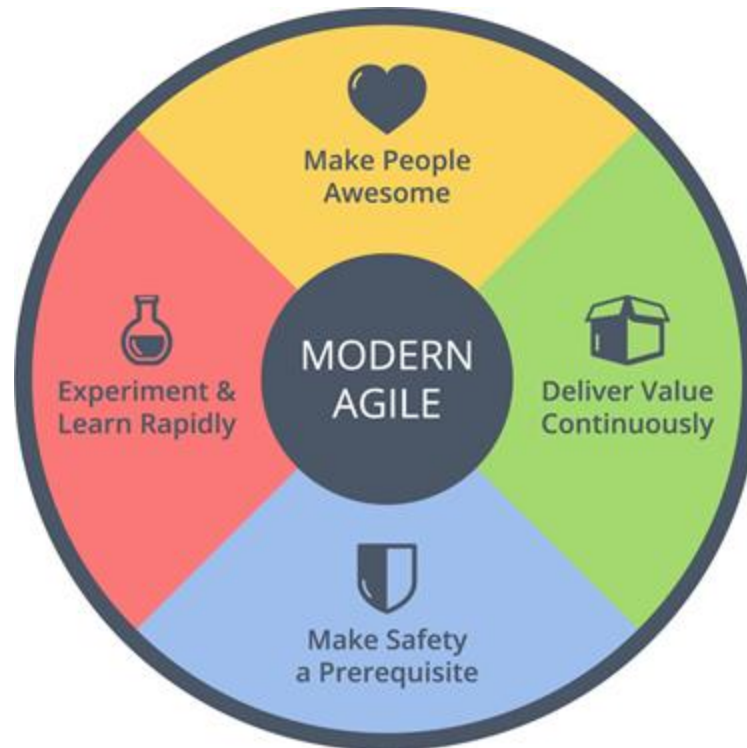
1.3 Describe the Agile methodologies

Experiment & Learn Rapidly

Is a guiding principle of Modern Agile because it protects us from wasting time and helps us discover success faster

Make Safety a Prerequisite

Means establishing safety before engaging in potentially hazardous work



“Make People Awesome”

If you make customers awesome, they tend to be natural promoters of your products or services

Deliver Value Continuously

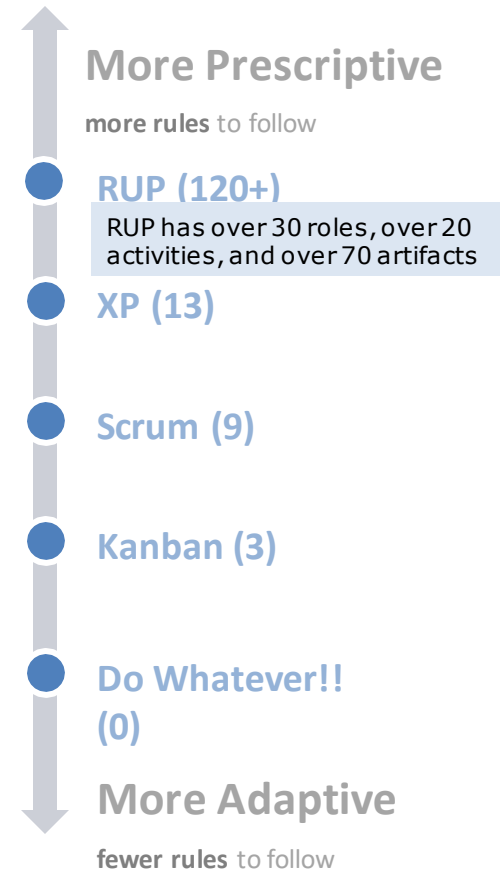
Anything valuable that hasn't been delivered isn't helping anyone. How might we deliver the right outcomes faster

Agile Umbrella

1.4 The great methodologies debate



* Check wikipedia for list of all Agile methods

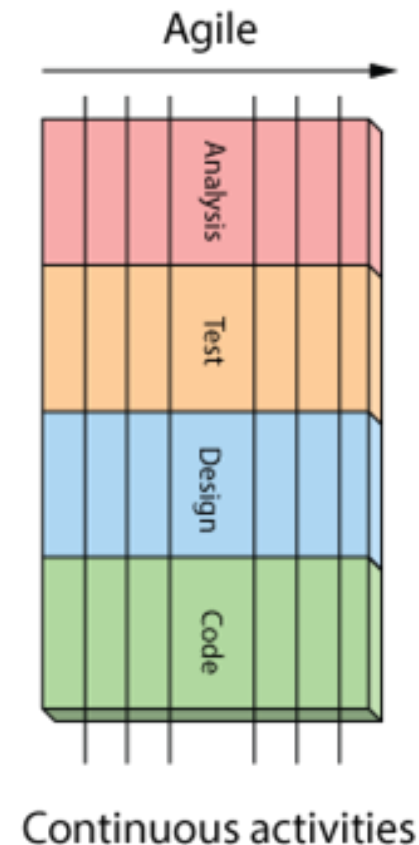
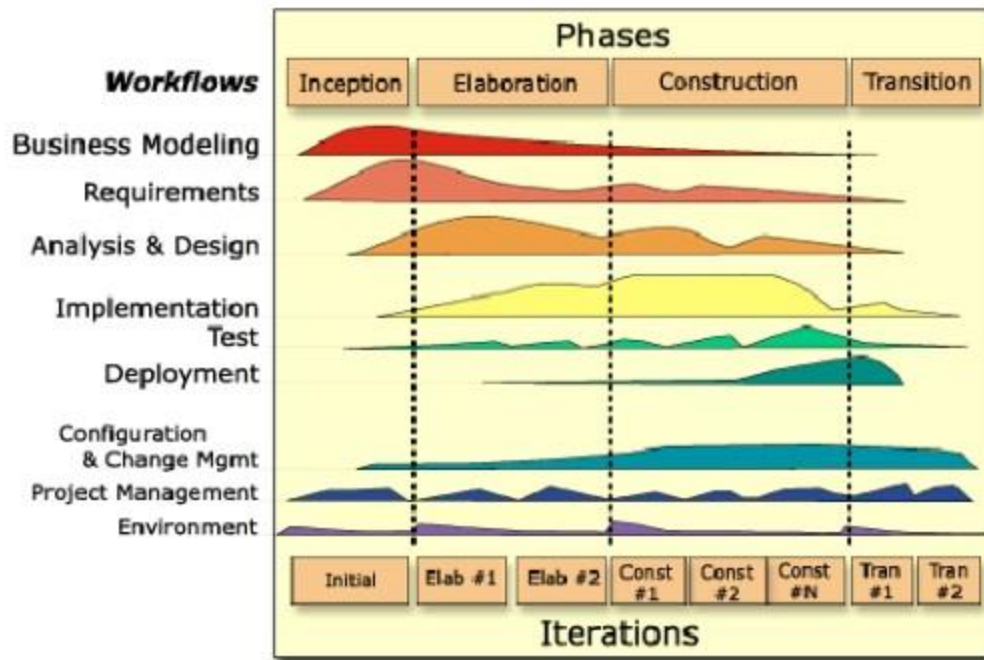


How is Agile different?

1.4 The great methodologies debate

The RUP has determined a project life-cycle consisting of four phases. These phases can break down into small iterations if needed

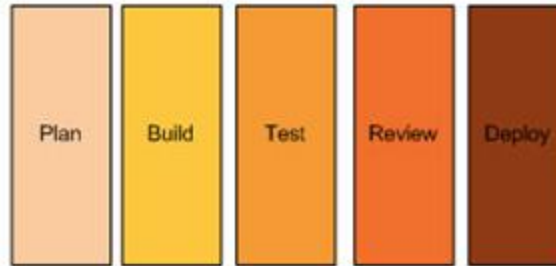
You are never done A&D, coding and testing on Agile project. So long as there are features to build, these activities continue for the duration of the project



RUP vs Agile: another view



RUP model



Agile model



Large Companies that use Agile...and lots of local guys, too



13,107

Companies using IBM Rational Unified Process



Project Management

IBM Rational Unified Process (14.76%)

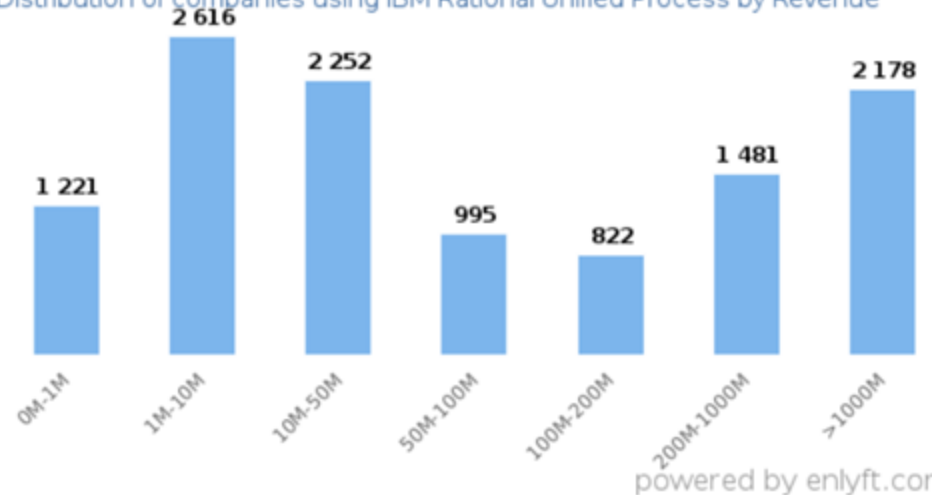
Asana (12.56%)

Smartsheet (11.17%)

Atlassian Greenhopper (6.06%)

[View other alternative products](#)

Distribution of companies using IBM Rational Unified Process by Revenue



Of all the customers that are using IBM Rational Unified Process, a majority (52%) are small (<\$50M), 31% are large (>\$1000M) and 9% are medium-sized.

Main characteristics comparison: RUP vs Agile

1.4 The great methodologies debate

RUP:

Heavyweight, Is a Framework

- Use-case driven from inception to deployment, small team size
- Architecture-centric, model visually, well-documented, lots of artifact
- Iterative and incremental, where large projects are divided into smaller projects
- Long-term detailed planning

Agile:

Lightweight, Like a Philosophy

- Small cross-functional teams are used
- Daily status meetings are held, less artifact can be better
- Short timeframe increments for each change, short-term planning
- A working project is completed at end of each iteration and demonstrated to stakeholders

Main problems comparison: RUP vs Agile

1.4 The great methodologies debate

RUP:

- It mostly **relies on the ability of experts** and professionals to assign the activities to individuals
- It is a rather complex method which **makes its implementation challenging**, particularly for smaller businesses, teams or projects
- A **long time is spent doing requirements** or design work before programming starts
- Too **many roles, artifacts** required
- **High cost** in managing, following RUP project
- Tool-set are useful but **expensive**

Agile:

- It can be difficult to keep the interest of **customers / users** who are involved in the process.
- Team members may be unsuited to the **intense involvement** that characterizes agile methods.
- **Prioritizing** changes can be difficult where there are **multiple stakeholders**.
- Maintaining **simplicity requires extra work**.
- **Contracts** may be a problem as with other approaches to iterative development.
- Because of their focus on small, tightly-integrated teams, there are problems in **scaling** agile methods to **large systems**.
- **Less** emphasis on **documentation** - harder to maintain when you get a new team for maintenance

Experts review ⁽¹⁾

1.4 The great methodologies debate

Quora     

Is the Rational Unified Process (RUP) widely used in the companies such as Facebook, Twitter, Google, MS or others? What kind of attitudes do the major IT companies hold toward it?

 Answer  Follow · 5  Request    

2 Answers

 **Bruce Taylor**, Aging as gracefully as I can.
Answered Nov 22, 2018

Managers worship software methodologies (like RUP).

Programmers find ways to subvert them in order to get the job done.

51 views · View 1 upvote

 1     

 [Add Comment](#)

Quora     

Methodology Scrum (product development) Agile Software Development

Project Management 

As an IT professional, which development methodology is more successful: RUP (Rational Unified Process) or Scrum?

 Answer  Follow · 2  Request    

2 Answers

 **Chuck Cobb**, Agile Project Mgt Author and Instructor (Over 100K Students)
Answered Nov 14, 2018

Scrum is many-times more widely-used than RUP. RUP was originally closely-coupled to the Rational tool set which was very expensive and difficult to learn and use. And, most of the Rational tools that were originally used by Rational in conjunction with RUP are no longer supported by Rational.

A number of variations of the unified process have evolved since then that are independent of the Rational tool set - for example there has been the Enterprise Unified Process, and Scott Ambler's Agile Modeling approach. All of those variations have very limited use today.

Based on the sheer number of proj... [\(more\)](#)

    1  

Experts review (2)

1.4 The great methodologies debate

 software and systems.

Answered Oct 2, 2014

First, you need to understand that, in my humble opinion, you should never use the Rational Unified Process (RUP) as it is out of the box. Like the direct adoption of any process, that is a recipe for disaster.

Whenever adopting any process, you need to figure out what you currently have - and it is OK to say you have nothing! Once you know that, you can start planning on where you want to be and how to get there.

RUP should be considered as a set of best practices with what I deem to be a sample process for the development of a large, complex system. It is unlikely that it will fit either where you... [\(more\)](#)

 7    2  

2 comments from Muhammad Zaheer and more



Madhu Siddalingaiah, worked at To be added later.....

Answered Oct 4, 2014

I'm a certified [Project Management Professional](#) , [Lean Six Sigma](#)  black belt, and I have led multiple companies through successful CMMI level 3 appraisal. In other words, I'm a certified process weenie. I know enough about process to annoy [Sheldon Cooper](#) .

I would use RUP only when required to do so, even then I would ask for a justification. There are many effective processes available to choose from today, RUP is just one of them.

At the end of the day, there is only one process that matters: the one that works for you and your organization. Whatever you decide, it is critical to evaluate effic... [\(more\)](#)



Andrew McGregor, former Site Reliability Engineer at Google (2013-2019)

Answered Jan 27, 2018

Many Agile practices have become standard nearly everywhere.

Google has a lot of them too: test-driven design, meticulous attention to test coverage, short release cycles. But not formal Agile methods in most places... except that implicitly, everything is driven by project plans chunked into bugs, so really it's Kanban by another name. Some teams explicitly do Kanban or Scrum.

Kanban works better for SRE teams than Scrum, because there are too many issues arising and random events to use a strict iteration process.

7.6K views · View 30 upvotes

 30   



Add a comment...

Add Comment



Alex Wiesen, worked at Google

Answered Mar 7, 2016 · Upvoted by [Sandra Liu Huang](#), worked at Google and [Marc Bodnick](#), Former institutional investor in Facebook

Google has a great deal of variety in terms of how teams work. From my experience Agile isn't that common for engineering teams, but it does get used occasionally. (In fact, one of the teams I manage is using some Agile development practices in their current project.)

In my previous life I spent many years creating project management software. My advice for people is to not focus too much on one approach to project management or development, and instead be open to different ideas and techniques. The best practices for software development are really unique to each team and each project -- there's no one size fits all solution out there.

10.7K views · View 22 upvotes

Experts review ⁽³⁾

1.4 The great methodologies debate

**In a nutshell, why do a lot of developers dislike Agile?
What are better project management paradigm
alternatives?**

 Answer  Follow · 353  Request  7  

100+ Answers



Miles English, I'm a programmer.

Answered Jul 5, 2016 · Upvoted by Jared Knipp, MS Software Engineering, Texas Tech University and Erdem Toraman, M.S Software Engineering, University of Tartu (2018)

Originally Answered: In a nutshell, why do a lot of developer dislike Agile?

This story is 100% true.

I got hired onto a team of construction workers to build a house. We set up a meeting with Management to find out what kind of house they wanted us to build, where's the floor plan, what it's going to be used for, who it's for, etc. Management said that they didn't know all that, we should just get started. They told us that we were going to use "Agile" which means that we just work on small deliverables and build the thing incrementally.

The developer team lead argued that we at least need to know how big the thing is going to be so that we can get started pouring the foundation, but Management told him they just don't know. "What we do know," Management said, "is that the house is going to have a bathroom. Just start there, and we'll know more when it's done. You have two weeks."

So we just bought a port-a-potty, and screwed around on the internet for two weeks. Management was outraged. "You call this a house? This is the worst house ever! It doesn't even have a tv!"

 3.7K   11  

Quora



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Software Development Methodologies Agile Software Development Software Engineering

Startup Strategy 

**Why does agile development fail in most startup
companies?**

 Answer  Follow · 3  Request   

6 Answers



Chuck Cobb, Agile Project Mgt Author and Instructor (Over 100K Students)

Answered Jun 27, 2019

Originally Answered: Why do companies struggle to adopt agile development?

- It's difficult and requires training. Many companies try to short-circuit that process and wind up doing it mechanically without fully understanding the principles and values behind it at a deeper level
- It can also require an enterprise-level transformation to break down organizational barriers for it to work effectively. It can also require some level of cultural change which can be a difficult thing to do
- In addition, Agile/Scrum is primarily designed around small, single-team projects, scaling it to large, complex enterprise-level projects can be a very challenging thing to do

177 views · View 2 upvotes · Answer requested by Allison Davis

Conclusion

- RUP will be main methodology in this course due to the following reasons:
 - It is a framework
 - Well documented, having templates, tools, well explanation
 - Model visually
 - Easy to learn, do not required much experience as Agile
 - Good method to learn OOP, OOD
 - Can be customized into small version with less artifacts